

Project Proposal: SkillSwap

Project Title

SkillSwap: A Peer-to-Peer Skill Exchange and Booking Platform

Project Type

Full-Stack Web Application (Marketplace and Scheduling System)

SkillSwap is a database-driven, multi-user web application that allows students to offer skills, discover tutors, and schedule learning sessions through a centralized platform.

High-Level Scope (In Scope)

The project will focus on developing a Minimum Viable Product (MVP) with the following core features:

1. User Authentication and Profiles

- User registration, login, and logout
- Secure password handling
- Personal profile pages with bio and listed skills
- Profile editing functionality

2. Skill Listings Management

- Create, view, edit, and delete skill listings
- Listings include skill name, description, tags, and availability
- View tutor profiles from listings

3. Search and Discovery

- Browse all available listings
- Keyword-based search

- Basic filtering by skill category

4. Booking and Scheduling System

- Request tutoring sessions
- Accept or reject booking requests
- Track booking status (Pending, Accepted, Rejected, Completed, Cancelled)
- Personal dashboard for managing bookings

5. Messaging System

- In-app messaging between tutors and learners after booking acceptance
- Message threads linked to bookings

6. Rating and Review System

- Learners can rate tutors after completed sessions
- Display average ratings on profiles and listings

7. Administration Tools

- Admin dashboard for managing users and listings
- Ability to remove inappropriate content or deactivate accounts

8. Skill Request Board (Reverse Marketplace)

- **Request Creation:** Students can post "Help Wanted" requests for skills not currently listed on the platform.
- **Application System:** Other users (potential tutors) can browse requests and apply to teach them by submitting a short pitch or qualification summary.
- **Applicant Selection:** The original requester can view a list of applicants and "Accept" a tutor, which automatically initiates the booking process.

Out of Scope

The following features are explicitly excluded to ensure the project remains achievable:

- Online payment processing
- Video conferencing and live tutoring
- Mobile application development
- AI-based recommendations
- External calendar integration

- SMS or email notification systems
 - Group sessions and subscription models
 - Advanced analytics and reporting
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Trello Board Link

<https://trello.com/b/HofdCj69/skillswap-swe-project>

The Trello board will be used for sprint planning, task assignment, and progress tracking. It will follow an Agile-style workflow with clearly defined stages.

Project Overview

1. Technology Stack

Backend:

- Python (Django Framework)

Database:

- MySQL

Frontend:

- Django Templates
- HTML, CSS
- Minimal JavaScript for user interactions

Authentication:

- Django Authentication System

Version Control & Tools:

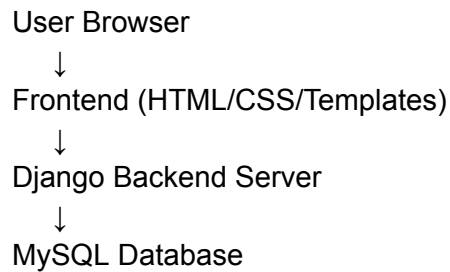
- GitHub
 - Trello
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2. System Architecture

SkillSwap will follow a client-server architecture:

1. Users interact with the system through a web browser.
2. The browser sends requests to the Django backend server.
3. The backend processes business logic and communicates with the MySQL database.
4. Data is returned to the frontend for display.

Architecture Flow:



3. Context Diagram Description

Actors:

- Learner (Student seeking skills)
- Tutor (Student offering skills)
- Administrator

System:

- SkillSwap Platform

Interactions:

- Users register and log in
 - Tutors create skill listings
 - Learners post skill requests and review applicants
 - Tutors apply to open skill requests
 - Learners search and book sessions
 - Users communicate through messaging
 - Learners submit reviews
 - Admin manages system content
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4. Database Design Overview

Key entities in the system include:

- **Users:** User information and authentication data
- **Listings:** Skills offered by tutors
- **SkillRequests:** "Help Wanted" posts created by learners seeking specific skills
- **Applications:** Records of tutors applying to fulfill a SkillRequest
- **Bookings:** Session requests and statuses
- **Messages:** Communication records
- **Reviews:** Ratings and feedback

These entities will be related using foreign keys to maintain data integrity.

5. Development Timeline

Week 1: Setup and Core Features

- Project setup and repository creation
- Database schema design
- User authentication implementation
- Skill listings CRUD functionality
- Basic frontend layout

Week 2: Booking, Messaging, and Reviews

- Booking workflow implementation
- Skill Request Board development (Post Request/Apply)
- Messaging system development
- Rating and review system
- User dashboards

Week 3 : Testing and Deployment

- System testing and bug fixing
 - Admin panel setup
 - UI improvements
 - Deployment preparation
 - Final demo preparation
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6. Trello Board Structure

Lists:

- Backlog
- To Do
- In Progress
- Testing
- Done

Example Tasks:

- Configure Django and MySQL
 - Implement authentication system
 - Design database models
 - Develop booking API
 - Create messaging interface
 - Implement review system
 - Perform system testing
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Conclusion

SkillSwap aims to provide a structured and accessible platform for students to exchange skills and knowledge. By focusing on essential marketplace/reverse marketplace and scheduling features, the project will deliver a functional and scalable MVP within the given timeline. The chosen technology stack ensures rapid development, maintainability, and alignment with software engineering best practices.